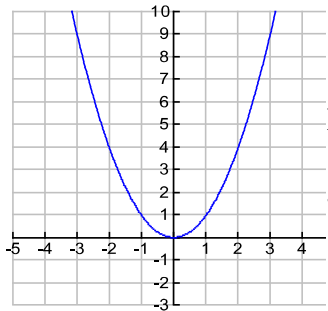
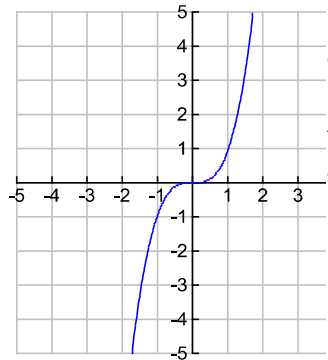
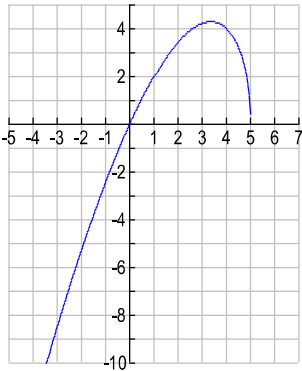
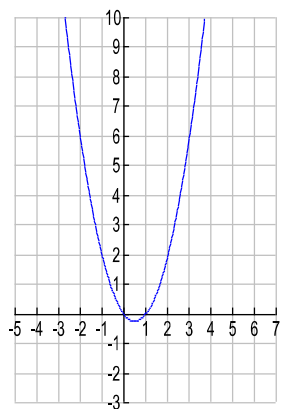


Practice:

1. Review your notes and the examples below. Then, for each of the practice questions:
- State in words how you know from the graph that the function is even, odd, or neither.
 - Prove algebraically that the function is even, odd, or neither. Use good form!
 - Complete the summary statements at the end.

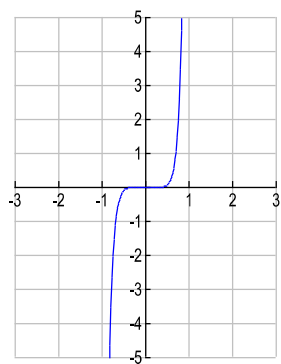
Function	Graph	Odd or even	Algebraic Proof:
$f(x) = x^2$	 Even since $f(x)$ is symmetrical about the y-axis.		$f(x) = x^2$ $f(-x) = (-x)^2$ $= x^2$ $= f(x)$ Therefore $f(x)$ is even.
$f(x) = x^3$	 Odd since $f(x)$ is symmetrical through the origin.		If $f(x)$ is odd then $f(-x) = -f(x)$ $f(-x) = (-x)^3$ $= -x^3$ $= -f(x)$ $\therefore f(x)$ is odd
$f(x) = x\sqrt{5-x}$	 Neither even nor odd since the function is not symmetrical through the origin or about the y-axis.		$f(x) = x\sqrt{5-x}$ Even? $f(-x) = (-x)\sqrt{5-(-x)}$ $= -x\sqrt{5+x}$ $\neq f(x)$ Therefore, $f(x)$ is not even. Odd? $f(-x) = (-x)\sqrt{5+(-x)}$ $= -x\sqrt{5-x}$ $\neq -f(x)$ $f(x)$ is neither even nor odd Hint: Always check if the function is even first. Then, check if it is odd.

$$f(x) = x^2 - x$$



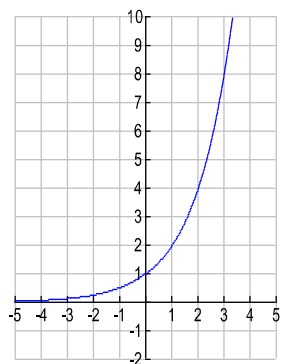
$$\begin{aligned} f(-x) &= x^2 + x \\ &= -(-x^2 - x) \\ &\text{neither} \end{aligned}$$

$$f(x) = (x + x^3)^5$$



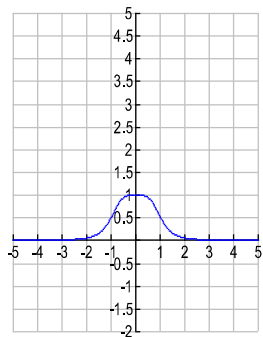
$$\begin{aligned} f(-x) &= (-x - x^3)^5 \\ &= \left[-(x + x^3)^5 \right] \\ &= -(x + x^3)^5 \\ &= -f(x) \\ &\text{Odd function} \end{aligned}$$

$$f(x) = 2^x$$



$$\begin{aligned} f(-x) &= 2^{-x} \\ &\text{neither} \end{aligned}$$

$$f(x) = \frac{1}{x^4 + 1}$$



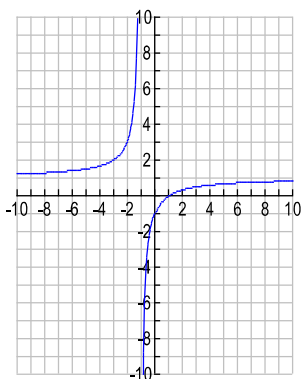
$$f(-x) = \frac{1}{(-x)^4 + 1}$$

$$= \frac{1}{x^4 + 1}$$

$$= f(x)$$

Even Function

$$f(x) = \frac{x-1}{x+1}$$



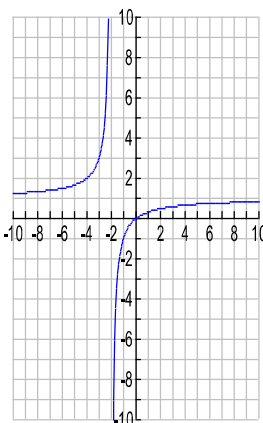
$$f(-x) = \frac{-x-1}{-x+1}$$

$$= \frac{-(x+1)}{-(x-1)}$$

$$= \frac{x+1}{x-1}$$

neither

$$f(x) = \frac{x}{x+2}$$



$$f(-x) = \frac{-x}{-x+2}$$

$$= \frac{x}{x-2}$$

neither

2. If $f(x)$ is an odd polynomial function, which of the following are odd polynomial functions?

- | | |
|-------------------|-------------------|
| i. $y = f(x+2)$ | i. <i>Not odd</i> |
| ii. $y = -f(x)$ | ii. <i>Odd</i> |
| iii. $y = f(2x)$ | iii. <i>Odd</i> |
| iv. $y = 2f(x)$ | iv. <i>Odd</i> |
| v. $y = f(x) + 2$ | v. <i>Not odd</i> |
| vi. $y = f(-x)$ | vi. <i>Odd</i> |

3. $y = f(x)$ is an even polynomial function, which of the functions in question 2 would be even polynomial functions?

- i. *Not even*
 ii. *Even*
 iii. *Even*
 iv. *Even*
 v. *Even*
 vi. *Even*

4. Determine an equation, in factored form, for an **even** polynomial function with a turning point at (2,0). Show algebraically that the function is even.

Answers may vary. One such equation is $y = (x-2)^2(x+2)^2$;

note that $f(-x) = (-x-2)^2(-x+2)^2 = (x+2)^2(x-2)^2 = f(x)$

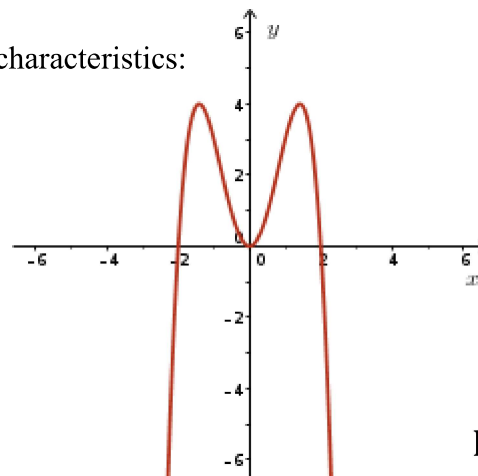
5. Determine an equation, in factored form, for an **odd** polynomial function with one of its zeros at $x=2$. Show algebraically that the function is odd.

Answers may vary. One such equation is $y = x(x-2)(x+2)$.

6. Sketch the graph of a polynomial function with the following characteristics:

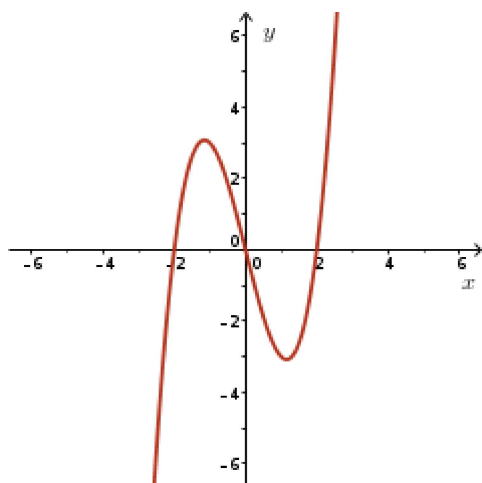
- the function is an even function
- as $x \rightarrow \infty, y \rightarrow -\infty$
- the function has exactly 3 x-intercepts

Answers may vary.

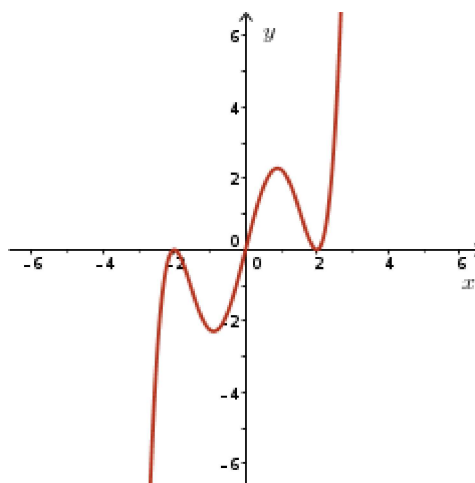


7. Sketch the graph of two polynomial functions of different degree with the following characteristics:
- the function is an odd function
 - as $x \rightarrow \infty, y \rightarrow \infty$
 - the function has exactly 3 x-intercepts

Answers may vary.



and



8. Determine, algebraically, if $f(x) = -2x(x-4)^2(x+4)^2$ is even, odd, or neither.

$$\begin{aligned} f(-x) &= -2(-x)(-x-4)^2(-x+4)^2 \\ &= 2x(x+4)^2(x-4)^2 \\ &= -f(x) \end{aligned}$$

Odd Function

9. Determine, algebraically, if $f(x) = (4x+8)(x-2)^2(2+x)$ is even, odd, or neither.

$$\begin{aligned} f(x) &= 4(x+2)(x-2)^2(2+x) \\ &= 4(x+2)^2(x-2)^2 \\ f(-x) &= 4(-x+2)^2(-x-2)^2 \\ &= 4(x-2)^2(x+2)^2 \\ &= f(x) \end{aligned}$$

Even Function

10. A quintic polynomial function that is classified as an **odd** function passes through the points $(2,0)$, $(3,210)$, $(4,0)$. Determine the equation of $f(x)$ in factored form.

$$f(x) = ax(x-2)(x+2)(x-4)(x+4)$$

$$f(x) = ax(x^2-4)(x^2-16)$$

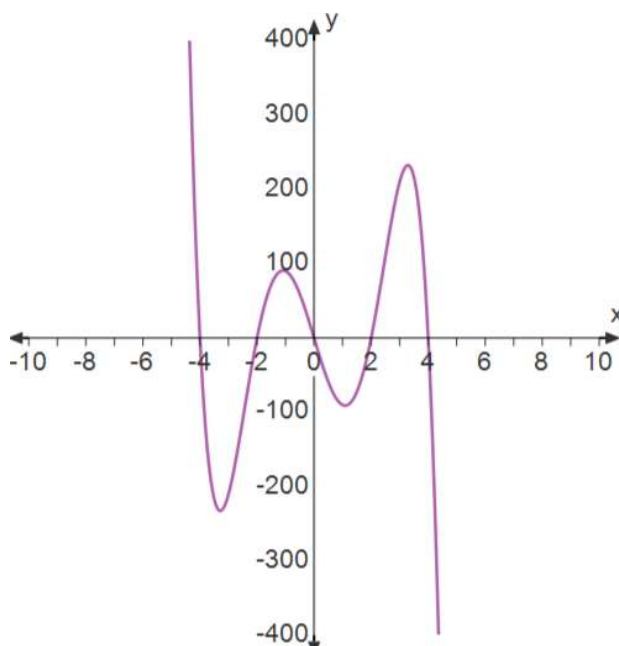
$$f(3) = 210$$

$$210 = 3a(3^2-4)(3^2-16)$$

$$210 = -105a$$

$$a = -2$$

$$\therefore f(x) = -2x(x^2-4)(x^2-16)$$



11. Given the function, $f(x)$, below is classified as an **even** polynomial function. Determine the exact value of $f(2)$. [Answer: $-\frac{25}{12}$]

$$f(x) = a(x+3)^2(x+1)(x-3)^2(x-1)$$

$$f(0) = \frac{9}{4}$$

$$\frac{9}{4} = a(0+3)^2(0+1)(0-3)^2(0-1)$$

$$\frac{9}{4} = -81a$$

$$a = -\frac{1}{36}$$

$$\therefore f(x) = -\frac{1}{36}(x+3)^2(x+1)(x-3)^2(x-1)$$

$$f(2) = -\frac{1}{36}(5)^2(3)(-1)^2(1)$$

$$\therefore f(2) = -\frac{25}{12}$$

